

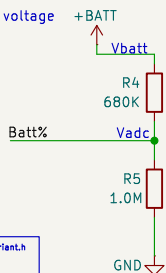
## Battery voltage divider for measuring voltage

$R4 = 470$   
 $R5 = 680$

# Voltage Division Equation  
 $V_{adc} = V_{batt} * (R5 / (R4 + R5))$

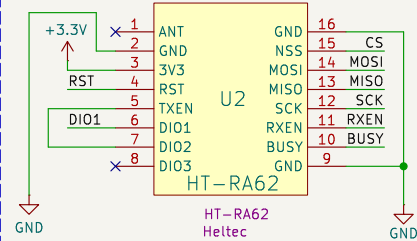
# Voltage Vbatt --> Vadc ratio  
 $ADC\_Division = V_{adc} / V_{batt} = (R5 / (R4 + R5))$   
 $R5 / (R4 + R5) = 0.591304347826087$

# Voltage Vadc --> Vbatt ratio  
 $ADC\_Multiplier = V_{adc} / V_{batt} = (R4 + R5) / R5$   
 $(R4 + R5) / R5 = 1.6913764705882353$

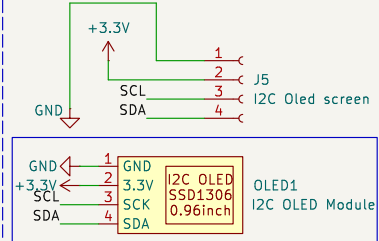


firmware/variants/nrf52840/dy/nrf52\_promicro\_dy\_tco/variant.h  
 #define VBAT\_DIVIDER\_COMP (1.73)  
 #define ADC\_MULTIPLIER VBAT\_DIVIDER\_COMP

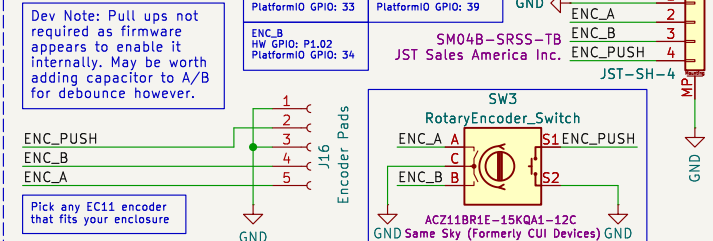
## HT-RA62 / RA-01SH



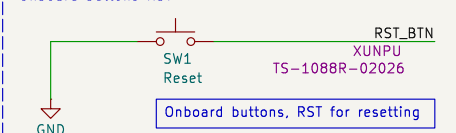
## Pins for connecting an I2C OLED screen



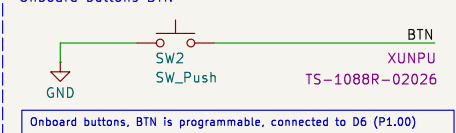
## Pads for connecting encoder



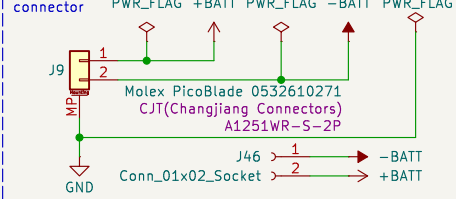
## Onboard buttons RST



## Onboard buttons BTN



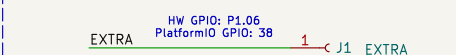
## Battery connector



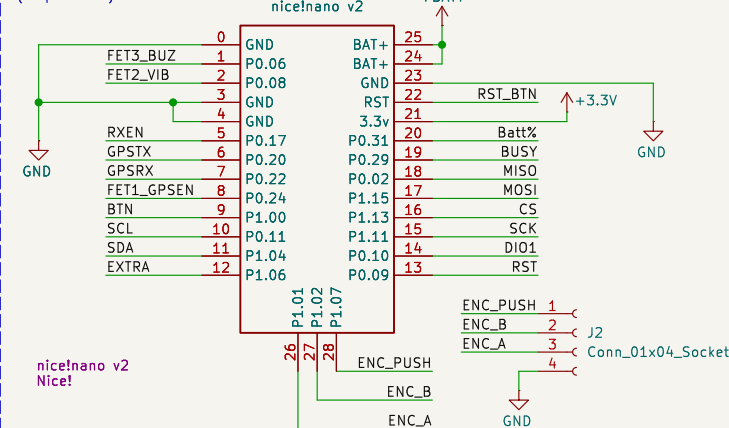
Molex PicoBlade 2-Pin  
 - MPC: Molex, MPN: 0532610271  
 - MPC: CJT (Changjiang Connectors), MPN: A1251WR-S-2P  
 - Pitch: 1.25mm  
 - Used By: HelTec LoRa 32  
 - This polarity is also common for lipos in Aliexpress

JST PH 2-Pin  
 - MPN: S2B-PH-SM4-TB  
 - Pitch: 2.00mm  
 - Used By: RAK19007 WisBlock  
 - This polarity is also common for lipos in Aliexpress

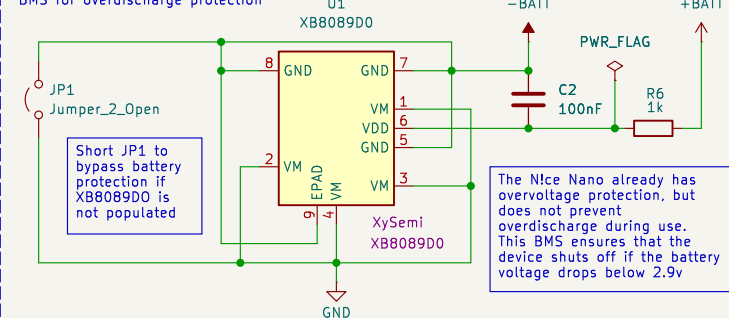
## EXTRA GPIO



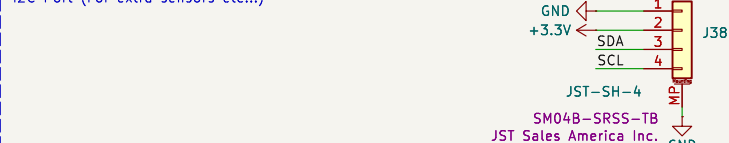
## Nice!Nano nrf52840 microcontroller (or promicro)



## BMS for overdischarge protection

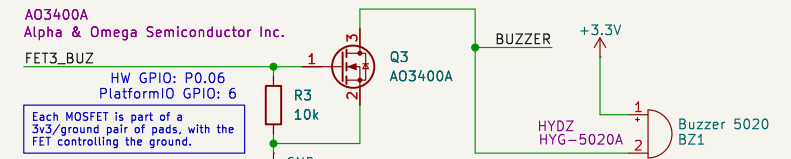
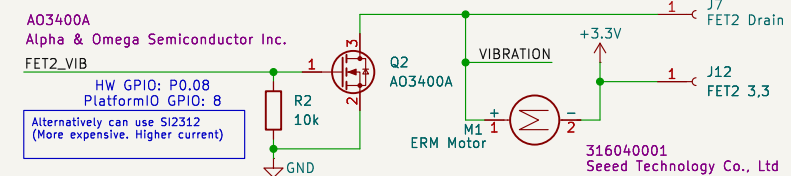
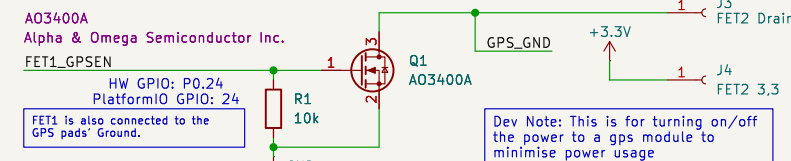


## I2C Port (For extra sensors etc...)

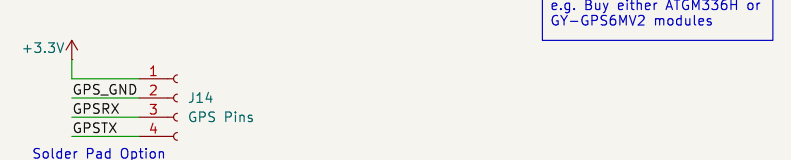


The JST-SH buddies  
<https://hackaday.com/2022/05/04/the-connector-zoo-i2c-ecosystems>

## Mosfets used for switching high current external peripheral



## GPS Module



Development Source: [https://github.com/mofosyne/roTec\\_pcb](https://github.com/mofosyne/roTec_pcb)  
 Fork Of FakeTec: [https://github.com/jtljac/fakeTec\\_pcb](https://github.com/jtljac/fakeTec_pcb)  
 Fork Of FakeTec: [https://github.com/gargomoma/fakeTec\\_pcb/](https://github.com/gargomoma/fakeTec_pcb/)

Sheet: /

File: fakeTecv5.kicad\_sch

Title: roTecTH Rev5.4

Size: A4 Date: 2025-12-26

KiCad E.D.A. 9.0.6

Rev: 5.4

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